Binary Additive Blends Including Pyridine Boron Trifluoride for Li-Ion Cells

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Pyridine Boron Trifluoride (PBF)-based electrolyte additive blends for $Li[Ni_{0.42}Mn_{0.42}Co_{0.16}]O_2$ (NMC442)/graphite pouch cells have been systematically investigated. Other additives investigated in the blends included methylene methane disulfonate (MMDS) and ethylene sulfate (DTD). Ultra-high precision coulometry (UHPC), electrochemical impedance spectroscopy (EIS), gas evolution and long-term cycling experiments were made. The results obtained in this work indicate that MMDS and DTD have good compatibility with PBF-type additives and some of their combinations showed significant improvements in coulombic efficiency, reducing impedance and maintaining good capacity retention during high-voltage cycling compared to PBF-type additives alone.
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Li-ion batteries (LIBs) are widely used in applications from portable consumer electronics to electric vehicles (EVs)[1–2] and high-voltage lithium ion batteries with higher energy density are a subject of much research. However, cells cycled under high-voltage conditions can suffer electrolyte oxidation which can cause impedance growth among other problems, thus causing capacity fading during cycling. Ma et al. reported impedance changes in NMC442/graphite Li-ion cells as a function of cycle number for cells tested in several different potential ranges with upper potentials from 4.2 V to 4.7 V.[3] Their results clearly indicated that, in order to improve capacity retention during high-voltage cycling, problems related to ever-increasing charge transfer impedance, R_{ct}, need to be overcome. Pyridine Boron Trifluoride (PBF) type additives are designed to control impedance growth due to the functionality of BF_3 group according to our previous study.[4] Furthermore, Wang et al.[5] and Nelson et al.[6] showed that, by combining multiple additives together, additional improvements to cell performance under challenging conditions could be obtained. Therefore, finding other additives which can also function as impedance-controlling agents and, when using together, are compatible with PBF-type additives is quite crucial.

Methylene methanedisulfonate (MMDS) is reported as one of the most promising high-voltage additives for lithium ion batteries with different positive electrode materials, such as $LiMn_2O_4$,[7] $LiNi_{0.5}Co_{0.2}Mn_{0.3}O_2$[8] and $LiCoO_2$.[9] The advantage of using MMDS as an additive becomes clear when charging to potentials higher than 4.2 V. MMDS apparently forms protective surface layers on the positive electrode surface and suppresses electrolyte decomposition.[10] MMDS also reduces the impedance as well as the self-discharge of NMC111/graphite cells during storage and cycling experiments.[11] Petibon et al. reported that other additives combined with MMDS yielded NMC 442/graphite pouch cells that maintained low impedance during cycling to high potential.[12]

In this work, several binary blends that combine PBF-type additives and MMDS or ethylene sulfate [1,3,2-Dioxathiolane-2,2-dioxide (DTD)] together were evaluated in EC-based electrolytes and NMC442/graphite pouch cells that were balanced to 4.7 V. The results obtained clearly demonstrate significant improvements in impedance control and cycling performance of cells cycled to high potentials of 4.4 or 4.5 V.

Pouch cells and electrolyte preparation.— 1 M $LiPF_6$ EC/EMC (3:7 wt% ratio, BASF) was used as the control electrolyte in the studies reported here. To this electrolyte, various synthesized PBF-type electrolyte additives shown in Figure 1 were added either singly or in combination with other additives. The purity of the additives was checked by NMR. NMR spectra are available in Figure S4 and in the supporting information of Ref. 4. Table I gives estimates of the purity of the synthesized materials used in this work based on the NMR spectra. Additive components were added at specified weight percentages in the electrolyte. Other standard electrolyte additives were also used for comparison. These included vinylene carbonate (VC, BASF, 99.97%), prop-1-ene,1,3-sultone (PES, Lianchuang Medicinal Chemistry Co., 98.20%), methane methylenedisulfonate (MMDS, Guangzhou Tinci Co. Ltd, 98.70%), tris trimethylsilyl phosphite (TTSPi, TCI America, >95%), triallyl phosphate (TAP, TCI America, >94.0%) and ethylene sulfate (DTD, Aldrich,98%). PBF-type additives were synthesized at Dalhousie University as previously described.[4]

Dry $Li[Ni_{0.42}Mn_{0.42}Co_{0.16}]O_2$ (NMC442)/graphite pouch cells (240 mAh) balanced for 4.7 V operation was obtained without electrolyte from Li-Fun Technology (Xinma Industry Zone, Golden Dragon Road, Tianyuan District, Zhuzhou City, Hunan Province, PRC, 412000, China). All pouch cells were vacuum sealed without

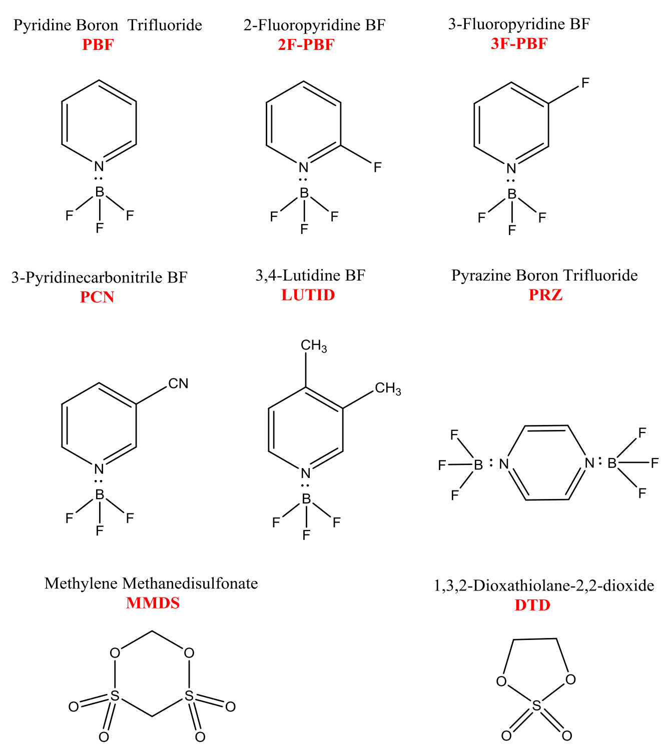

Figure 1. (a) Molecular formuli and abbreviations for the PBF-series additives, Methylene Methane disulfonate (MMDS) and 1,3,2-Dioxathiolane-2,2-dioxide (DTD).

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Table I. Purity of PBF and PBF-derivatives used as additives in electrolytes.

Product (electrolyte additive)	Short name	Purity (based on NMR)
Pyridine boron trifluoride	**PBF**	> 99%
3-fluoro pyridine boron trifluoride	**3F-PBF**	> 99%
2-fluoro pyridine boron trifluoride	**2F-PBF**	> 98%
3,4 lutidene pyridine boron trifluoride	**LUTID**	> 99%
3-pyridine carbonitrile pyridine boron trifluoride	**PCN**	> 99%
Pyrazine boron trifluoride	**PRZ**	> 99%

electrolyte in China and then shipped to our laboratory in Canada. Before electrolyte filling, the cells were cut just below the heat seal and dried at 80°C under vacuum for 14 h to remove any residual water. Then the cells were transferred immediately to an argon-filled glove box for filling and vacuum sealing. The NMC/graphite pouch cells were filled with 0.9 g of electrolyte. After filling, cells were vacuum-sealed with a compact vacuum sealer (MSK-115A, MTI Corp.). First, cells were placed in a temperature box at 40 ± 0.1°C where they were held at 1.5 V for 24 hours, to allow for the completion of wetting. Then, cells were charged at 11 mA (C/20) to 3.5 V. After this step, cells were transferred and moved into the glove box, cut open to release gas generated and then vacuum sealed again. After degassing, impedance spectra of the cells were measured at 3.8 V as described below. The NMC (442)/graphite cells destined for 4.4 or 4.5 V operation were degassed a second time at 4.5 V. The amounts of gas created during formation to 3.5 V and between 3.5 V and 4.5 V were measured and recorded.

Electrochemical impedance spectroscopy.— Electrochemical impedance spectroscopy (EIS) measurements were conducted on NMC/Graphite pouch cells before and after storage and also after cycling on the UHPC. Cells were charged or discharged to 3.80 V before they were moved to a 10 ± 0.1°C temperature box. AC impedance spectra were collected with ten points per decade from 100 kHz to 10 mHz with a signal amplitude of 10 mV at 10. ± 0.1°C. A Biologic VMP-3 was used to collect this data.

Ultrahigh precision cycling.— The cycling/storage procedure was carried out using the Ultra High Precision Charger (UHPC) at Dalhousie University.[13] Testing was between between 2.8 and 4.4 V at 40 ± 0.1°C. All cells were tested in temperature controlled boxes where the accuracy of the temperature was ± 1°C but the stability of the set temperature was ± 0.1°C. Therefore the temperatures are written as XY ± 0.Z°C in this work, where XY has two significant digits. If XY°C is to represent 10.°C, the decimal point must be present to indicate two significant figures. Cells were first charged to 4.400 V using currents corresponding to C/10, stored open circuit at 4.400 V for 20.00 h and then discharged to 2.800 V using currents corresponding to C/10. This process was repeated on the UHPC for 17 cycles. The cycling/storage procedure was designed so that the cells were exposed to higher potentials for significant fractions of their testing time. All pouch cells were cycled without clamps.

Long term cycling was conducted with upper cutoff potentials of 4.5 V. Two different temperatures and protocols were used for long term cycling. In the first protocol the cells were continuously charged and discharged at 100 mA between 2.8 – 4.5 V at 40. ± 0.1°C. Every 50 cycles included a C/20 cycle. In the second protocol the cells were continuously charged and discharged at 80 mA between 3.0 and 4.5 V at 55. ± 0.1°C. Cells were tested using a Neware (Shenzhen, China) charger system in both protocols.

Top of charge hold cycle experiments and frequency response analysis.— Some cells were tested extremely aggressively to really push the limits of the electrolyte. These cells were subjected to two test protocols:

Type 1 - The cells were charged and discharged at 80 mA current between 2.8 and 4.4 V and held at 4.4 V for 20 hours at 40. ± 0.1°C before discharging again on a Maccor series 4000 cycler;

Type 2 - the cells were charged and discharged at 44 mA between 2.8 and 4.4 V and held at 4.4 V for 20 hours every cycle. Automated impedance spectroscopy measurements were made after every three charge-hold-discharge cycles using a frequency response analyzer (FRA)/cycler system built at Dalhousie University. During FRA measurements, the cells were charged and discharged at 12 mA (C/20) and AC impedance spectra were collected every 0.1 V. In this paper, the (combined) diameter of the semicircle(s) in the Nyquist plot is called the charge transfer resistance, R_{ct}, and represents contributions from the transport of ions from the electrolyte through the SEI layer to the active material at both electrodes.

Determination of volume of gas evolved in pouch cells.— Ex situ (static) gas measurements were used to measure gas evolution during formation and during cycling.[14] The measurements were made using Archimedes' principle with cells suspended from a balance while submerged in liquid. The changes in the weight of the cell suspended in fluid, before and after testing are directly related to the volume changes by the change in the buoyant force. The change in mass of a cell, Δm, suspended in a fluid of density, ρ, is related to the change in cell volume, Δv, by

$$\Delta v = -\Delta m/\rho \tag{1}$$

Ex-situ measurements were made by suspending pouch cells from a fine wire "hook" attached under a Shimadzu balance (AUW200D). The pouch cells were immersed in a beaker of de-ionized "nanopure" water (18.2 MΩ · cm) that was at 20 ± 1°C for measurement.

Results and Discussion

Figures 2a and 2b shows the cell terminal voltage as the function of capacity during the first charge to the first degassing point

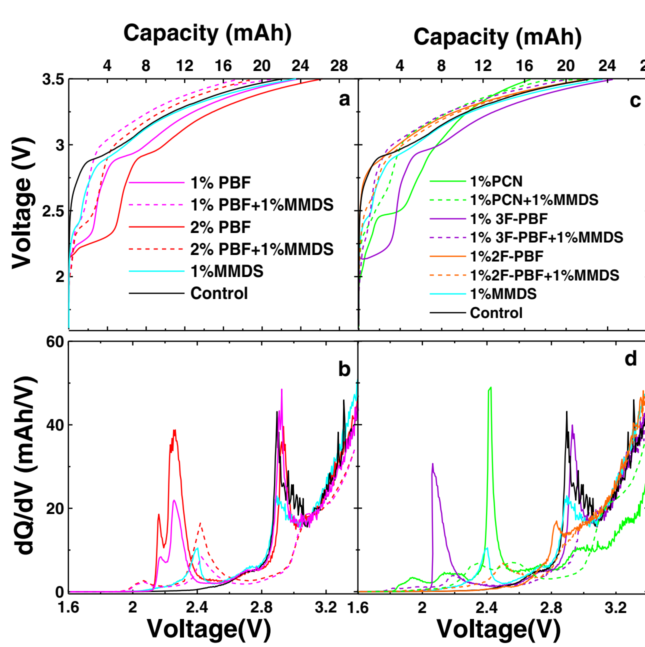

Figure 2. (a) The cell terminal voltage as a function of capacity during the initial charge (formation cycle) to the first degassing point (3.5 V) and (b) differential capacity (dQ/dV) versus voltage (V) during the same formation process for NMC442/graphite pouch cells containing different concentrations of PBF with 1%MMDS. (c) The cell terminal voltage as a function of capacity during the initial charge (formation cycle) to the first degassing point (3.5 V) and (d) differential capacity (dQ/dV) versus voltage (V) during the same formation process for NMC442/graphite pouch cells containing different combinations of PBF-derivative additives with 1%MMDS.

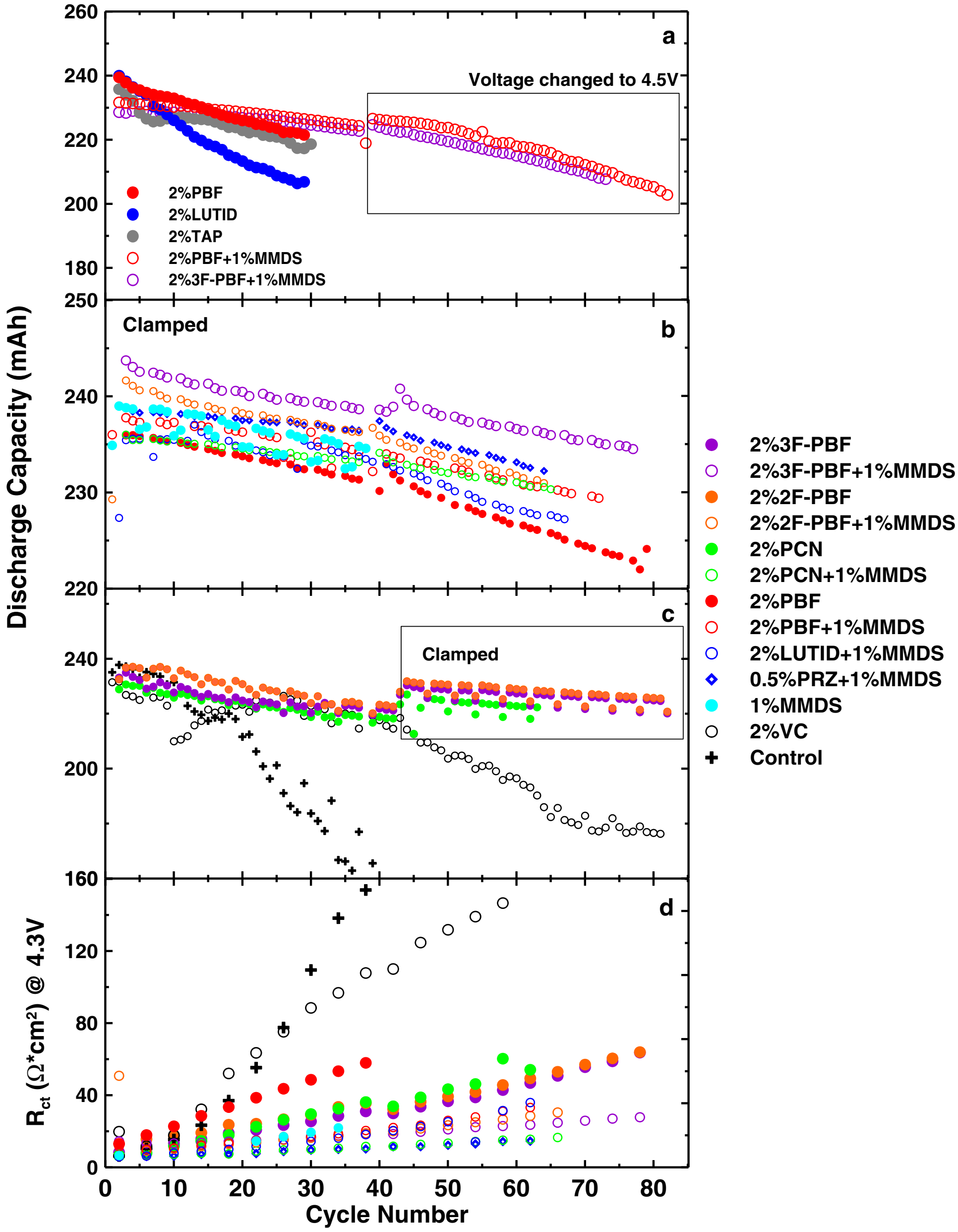

Figure 3. (a) Discharge capacity versus cycle number for NMC442/graphite cells cycled with type 1 protocol (– 20 h TOC hold at 4.4 V) using C/3 rate cycling at 40°C. (b) Discharge capacity versus cycle number for NMC442/graphite cells cycled with type 2 protocol FRA cycling at 40°C (C/5 with a 20 h hold at 4.4 V at top of charge). The cells were cycled with clamps from the beginning. (c) Data for comparative cells which did not include MMDS. Notice that the vertical scales in panels (b) and (c) are not the same. (d) R_{ct} measured at 4.3 V vs. cycle number for the cells described on (b) and (c) cells with different electrolytes.

(3.5 V) along with the differential capacity versus voltage for NMC442/graphite cells with different concentrations of the additives: PBF alone and PBF + MMDS combinations. Figure 2a shows that a plateau appeared at 2.3 V in PBF-containing cells (negative electrode potential ≈1.2 V vs Li/Li^+) and that the capacity of the plateau increased with PBF content. When 1 wt% MMDS was used in combination with PBF, the length of plateaus decreased and the peaks shifted to 2.4 V. Figure 2b shows the peak in dQ/dV vs. V corresponding to the plateau at 2.3 V is associated with the reduction of PBF at the graphite surface.[4] When 1%MMDS is added to PBF, the intensity of the peaks in dQ/dV decreased significantly compared to cells with control electrolyte or cells with PBF(only) additive. The peak in dQ/dV corresponding to the reduction of EC at 2.9 V (≈0.8 V vs Li/Li^+) disappeared when PBF + MMDS were used suggesting that the graphite surfaces were passivated predominantly by products from the reduction reactions of PBF and MMDS, not EC. Surface science studies are required to probe the effect of the simultaneous addition of PBF and MMDS.

Figure 2c shows the cell terminal voltage versus capacity during the first charge to the first degassing point (3.5 V) for NMC442/graphite pouch cells containing different PBF-type additives with and without 1% MMDS. Figure 2d shows that generally the peaks in dQ/dV vs V from the PBF-type additives appeared before EC reduction. Additionally, adding MMDS reduced the intensity of the peaks from the PBF-type additives and virtually eliminated the EC reduction peak at 2.9 V. Thus, different from the cells with only PBF-type additives or with MMDS alone, the combination of PBF-type additives with MMDS apparently can strongly reduce the amount of EC reduced during the formation cycle.

Figure 3a shows the charge-hold-discharge cycling results on NMC442/graphite cells operating at high potential and at 40°C. Type I cycle-hold-cycle protocols, where cells were held at 4.4 V for 20 h be-

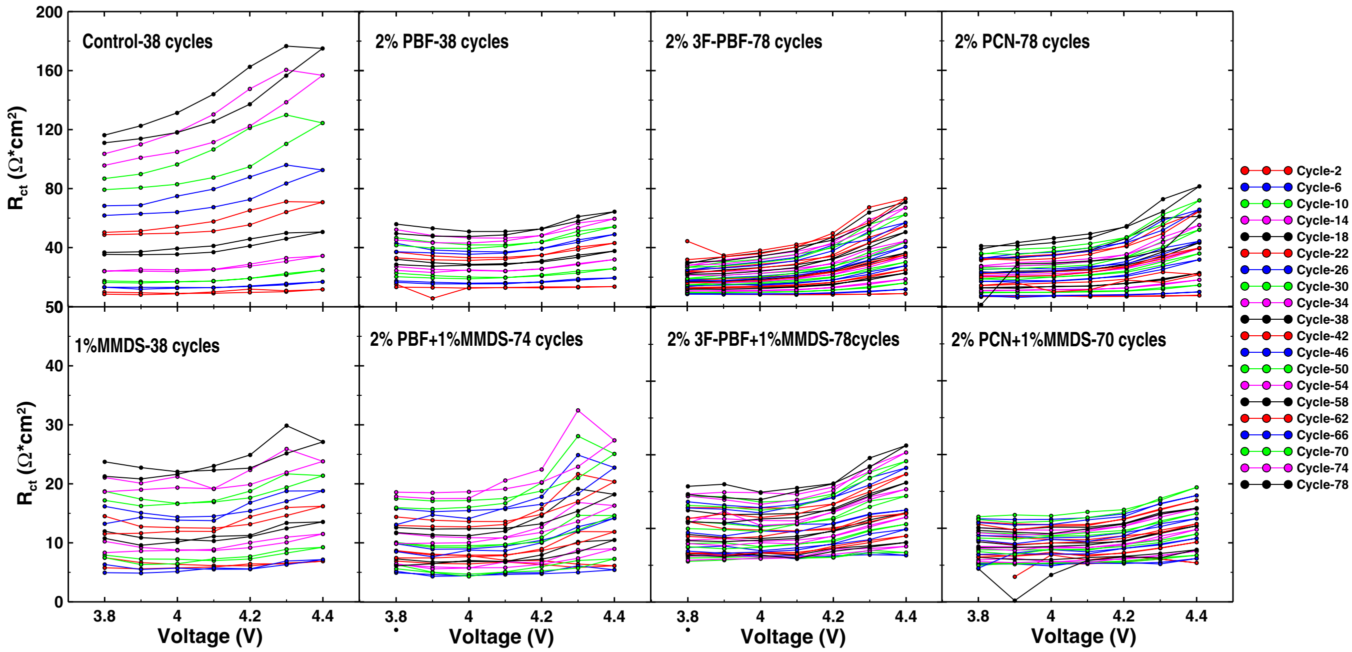

Figure 4. The charge transfer resistance, R_{ct}, as a function of voltage measured every 3 cycles for some of the cells in Figure 3: (Top Row) Results for cells with control electrolyte and PBF-type additives cells; (Bottom Row) Results for cells with 1% MMDS added to the electrolyte for comparison to the results in the panel directly above.

fore discharging, were initially used as described in the Experimental section. As discussed previously,[4] the PBF-type additives showed significant advantages when using in charge-discharge cycling to high potential. In order to further improve the high voltage performance, these PBF-type + MMDS additive combinations were tested by the same protocols in NMC442/graphite pouch cells. Figure 3a shows that cells with 2%PBF + 1%MMDS or 2%3F-PBF + 1%MMDS have better capacity retention compared to the cells with PBF-type additives alone. After 36 cycles, the cells with 2%PBF + 1%MMDS or 2%3F-PBF + 1%MMDS were charged and held at 4.5 V which induced increased capacity loss per cycle. In order to probe the impact of the combination of PBF-type additives and MMDS on the cell impedance, NMC442/graphite cells were tested using the type-2 protocol at 40°C.

Figure 3b shows the results for cells with six different PBF-type +MMDS blends and 1%MMDS only. Figure 3c includes our previous data[4] for cells with only PBF-type additives and for comparative cells with 2% VC and control electrolyte. In Figure 3b, all cells were charged and discharged with clamps[6] to force any generated gases to the edge of the pouch and maintain stack pressure. In Figure 3c, where results for cells with only single additives are reported, testing was made without clamps until clamps were applied to cells with good capacity retention at cycle 42. Figure 3c shows that the capacity retentions cells with PBF-type additives are much better than cells with control electrolyte and cells with 2% VC which show dramatic capacity loss after 20 and 45 cycles, respectively. Comparing data for cells with PBF-type additives (Figure 3c) with those containing both PBF-type and 1% MMDS (Figure 3b), (even after clamps are applied), the cells with both PBF-type and 1% MMDS additives do display slightly better capacity retention. Figure 3d shows the diameter of the semicircle of the impedance spectrum, called R_{ct} here, measured at 4.3 V plotted versus cycle number for all the cells described by Figures 3b and 3c. It is clear that the cells with PBF-type + 1%MMDS additive combinations showed the smallest impedance increase compared to cells with all other single additives.

Figure 4 shows the charge transfer resistance, R_{ct} measured during cycling from 3.8 to 4.4 V every 0.1 V and every 4 cycles for some of the cells from Figures 3b and 3c. Comparisons were made between cells with PBF-type additives and cells with PBF-type + 1% MMDS. Results for cells with control electrolyte and cells with 1% MMDS additive are included in Figure 4 as well. Figure 4 shows the large impedance increase with cycle number for the control cells. Both MMDS and the PBF additives alone reduce the rate of impedance increase with cycle number substantially. And 1%MMDS cells have much smaller impedance growth compared to the other single additive contained cells, which supported the reason by adding MMDS to reduce the impedance growth of PBF-type cells. So the best results are obtained for cells that contain both PBF-type + 1% MMDS additives. The impedance is remarkably stable both with potential and cycle number for the cell with 2% PCN + 1% MMDS.

In order to better understand the impedance growth observed during charge-discharge cycling in Figures 3 and 4, impedance spectra measured at each potential are displayed for cells with control electrolyte, with 2% 3F-PBF and with 2% 3F-PBF+1%MMDS at both cycle 2 and cycle 38 in Figure 5. Figures 5a and 5b show that the impedance of cells with control electrolyte increases immensely, especially at high potential, between cycle 2 and cycle 38, mainly through the growth of a low frequency semicircle. Our previous report[12] showed, using experiments on symmetric cells, that this impedance growth originated at the positive electrode side. Figures 5c and 5d show that this effect is strongly mitigated by the addition of 2% 3F PBF, but the growth of a low frequency semi-circle is still observed. One action of PBF-type additives is believed to be the stripping of LiF-buildup on the positive electrode through the formation of $LiBF_4$, which may be responsible for this effect.[8,15] Figures 5e and 5f show that cells with 2% 3F-PBF + 1% MMDS have even smaller impedance growth and the low frequency semicircle is virtually eliminated except at the highest potentials at cycle 38. Figure 5 clearly shows the impact of adding both PBF-type and MMDS additives to the same cell. Similar results were observed for cells with all PBF-type additives and all PBF-type + 1%MMDS combinations which are summarized in Figures S1, S2, S3 in the supporting information. This suggests that MMDS has good compatibility with PBF-type additives and by adding MMDS, the impedance growth especially at high voltage, is significantly reduced.

Ultra-high precision coulometry measurements were made on cells with PBF-type and PBF-type + 1% MMDS additives. UHPC measurements are difficult when there is a voltage hold segment because

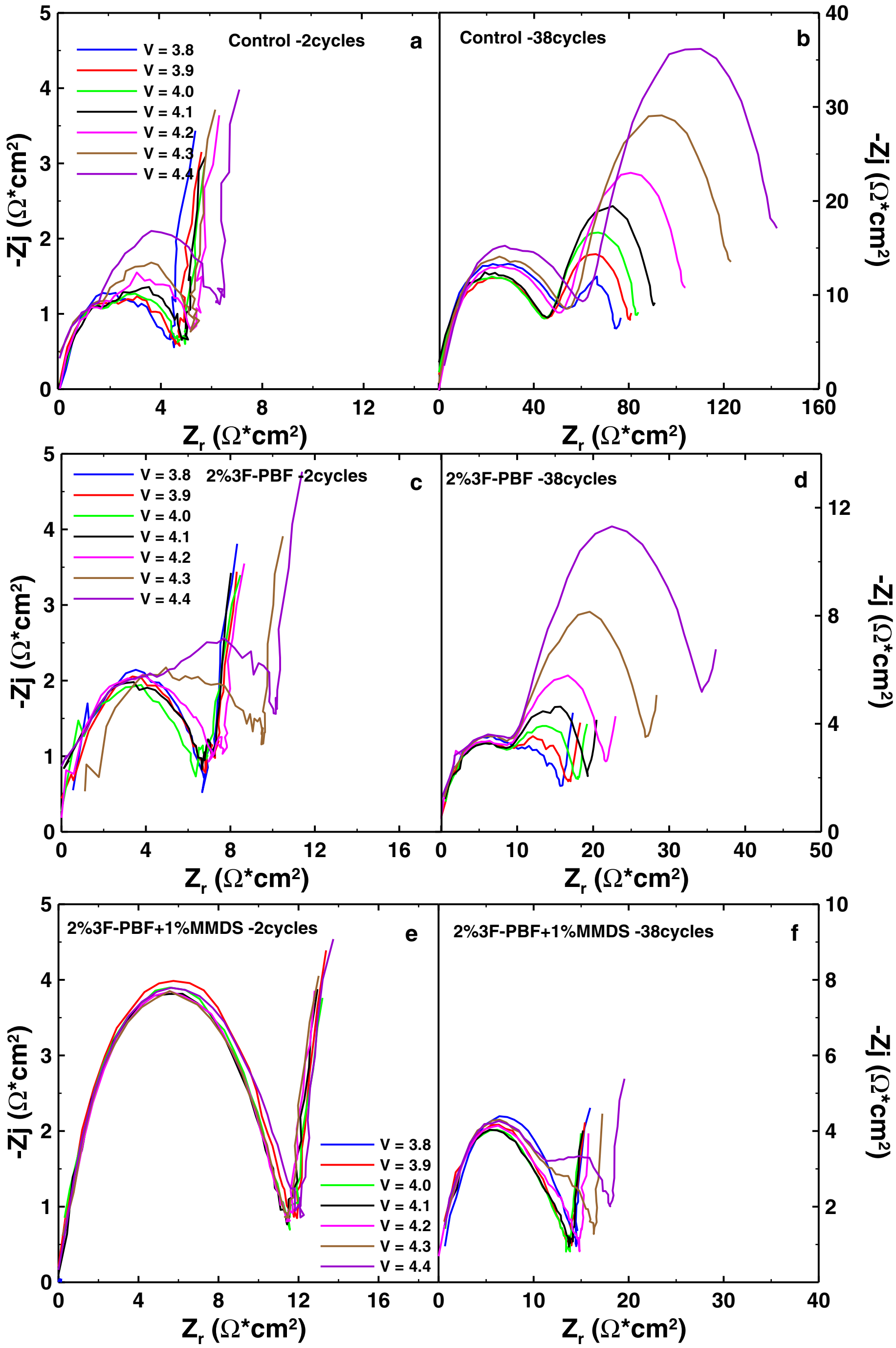

Figure 5. Comparison of the Nyquist plots of cells described by Figure 4 at various potentials and cycle numbers: a) Control electrolyte at cycle 2; b) control electrolyte at cycle 38; c) 2% 3F-PBF containing cells at cycle 2; d) 2% 3F-PBF containing cells at cycle 2; e) cells with 2% 3F-PBF + 1%MMDS at cycle 2; f) cells with 2% 3F-PBF + 1%MMDS at cycle 38.

it is hard to maintain high coulombic efficiency accuracy when the current varies substantially. Therefore, open circuit periods at the top of charge were used as a substitute. Figure 6 shows a schematic of the protocol used. The coulombic efficiency, discharge capacity, charge end point capacity, and the voltage drop during the 20 h OCV period were all measured versus cycle number.

Figure 7 shows the UHPC results for cells with five different additive combinations as well as cells with "PES211" and control cells which were charged and discharged according to the protocol in Figure 6. All these cells were cycled without clamps. Figure 7a shows the CE versus cycle number. Control cells showed very low CE and the noise in the data is thought to arise from gas generated due to severe electrolyte oxidation when the cells were exposed to high potential for a long time. Cells with PES211 and cells with 1% MMDS showed much better CE than control cells. The cells with PBF-type + 1% MMDS combinations showed similar CE values to those of the PES211 cells. Cells with 0.5% PRZ + 1%MMDS and cells with 1% 2F-PBF+1% MMDS have even higher CE than cells containing PES211 and MMDS alone. The noise in the CE versus cycle number data is much worse than expected based on the specifications of the UHPC used in these experiments. The noise originates due to stack pressure loss due to gas generation in these cells and can be mostly eliminated by clamping cells during testing.

Figures 7b–7e show: the discharge capacity, V_{drop} during the 20 h OCV period, ΔV (the difference between average charge and discharge potential) and the charge end point capacity all plotted versus cycle number. Figure 7b shows that all cells have similar capacity retention except for cells with control electrolyte. However, the results in Figures 7c–7e can be used to distinguish better between the various additives. The voltage drop during the

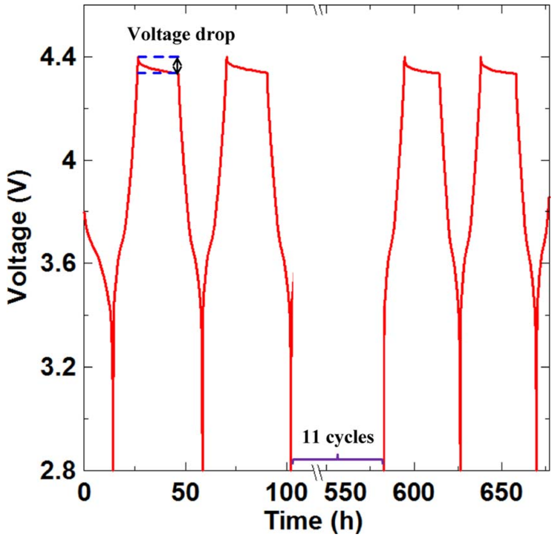

Figure 6. Schematic of the testing method used for the cycle/store protocol on the UHPC.

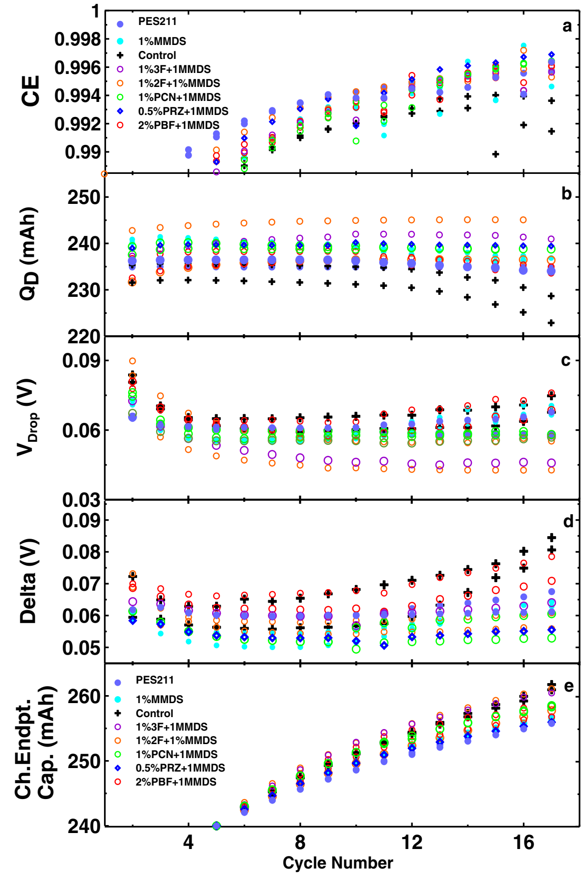

Figure 7. UHPC data for NMC442/graphite pouch cells with PBF-type + MMDS combinations. The cells were tested at 40°C with an upper cutoff of 4.4 V. (a) Coulombic efficiency (CE), (b) discharge capacity, (c) V_{drop} during the 20 h storage period after each charge and (d) ΔV, the difference between average charge and discharge potentials and (e) charge end point capacity. All are plotted versus cycle number.

OCV period is caused by electrolyte oxidation, transition metal ion dissolutions and shuttle reactions, etc.[16] and also includes the elimination of the IR overpotential when the cells are stopped charging. Figure 7c shows that the voltage drop in the cells with additives such as 0.5% PRZ + 1% MMDS, 1% 2F-PBF + 1% MMDS, 1% 3F-PBF + 1%MMDS and 1% PCN + 1%MMDS did not increase with cycle number, while it did for cells with control, 1% MMDS, PES211 and 2% PBF + 1% MMDS. Figure 7d mirrors the results in Figure 7c, suggesting impedance increase, not a fundamental change in the electrolyte oxidation rate, is the reason for the variation in Figure 7c. Figure 7e shows the charge end point capacity plotted versus cycle number for the various cells. Interpreting charge end point capacity slippage is complicated when cell impedance is not fixed during the cycling. As impedance increases, cells hit the upper cutoff potential earlier each cycle and charge end point capacity is measured as artificially small. As Figures 7c and 7d suggest, not all cells had constant impedance because some show increased V_{drop} and ΔV with cycle number. Therefore Figure 7e can be used to state with certainty that cells with 0.5% PRZ + 1% MMDS have excellent charge end point capacity slippage since these cells show invariant V_{drop} and ΔV in Figures 7c and 7d. However, none of the additive blends shows a dramatic decrease (i.e. 5x) in charge end point capacity slippage compared to control electrolyte, suggesting that electrolyte oxidation is still problematic (for 4.4 V cycling) for all these systems. Thus by adding 1%MMDS to PBF-type additives, the additives blends do not show dramatic effects in preventing electrolyte oxidation compared to PES211 or 1% MMDS alone. The major function of MMDS appears to be an impedance reducer for these combinations.

Figures 8a and 8b show the Nyquist plots of the impedance spectra of the pouch cells described by Figure 7 before and after UHPC

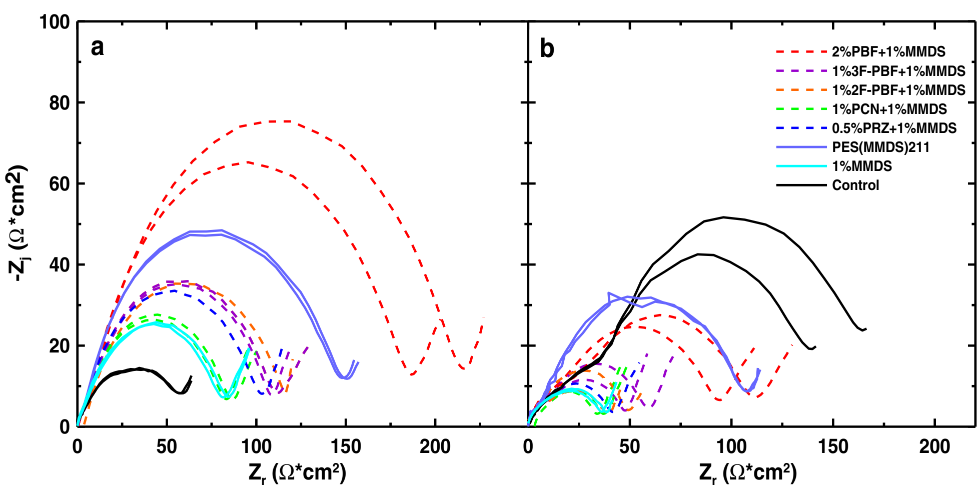

Figure 8. Impedance spectra of the cell in Figure 7; (a) before UHPC, (b) after UHPC.

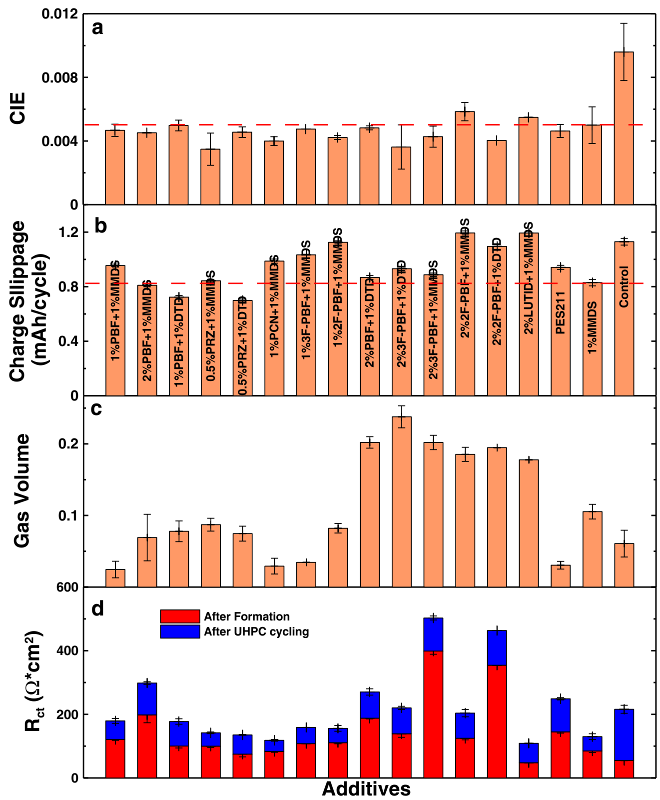

Figure 9. Summary of UHPC data for all the PBF-type + MMDS/DTD additive containing cells. (a) CIE, (b) charge end point capacity slippage calculated from last 6 cycles and (c) gas evolution. (d) Summary of R_{ct} data for the cells with PBF-type + MMDS/DTD additives and for comparative cells before and after UHPC testing. Error bars represent the standard deviation of the data.

cycling at 40°C. All impedance measurements were made at 3.8 V and at 10 ± 0.1°C. The diameter of the semi-circle in the impedance spectrum represents the sum of the resistive parts of the charge transfer impedance and the transfer of ions through the SEI layers for both electrodes. Figure 8a shows that the cells with combinations 2%PBF + 1%MMDS have a larger resistance after formation compared to cells with control, 1%MMDS or PES211. But after ∼750 hrs of charge/store UHPC cycling (protocol shown in Figure 6), the impedance of cells with PBF-type + 1% MMDS additive blends was dramatically reduced. Other comparative cells containing 1%MMDS or PES211 also showed an impedance decrease that suggests MMDS is useful in reducing the impedance.

Figure 9 summarizes the UHPC data collected from all the cells with PBF-type additives as well as cells with PBF-type + MMDS and cells with PBF-type + DTD. Figures 9a–9c show coulombic inefficiency (1-CE = CIE), charge end point capacity slippage and volume change of the cells after the charge/store cycling experiment. The CIE should be minimized for the best cells. A dashed line has been included in Figures 9a and 9b to indicate the level for cells with 1% MMDS. Most cells with PBF-type + MMDS and PBF-type + DTD have smaller CIE than control cells and cells with 1% MMDS. The charge end point capacity slippage was from the slope of the data in Figure 7e for the last 6 cycles. Using the charge/store protocol, the slippage measured for these cells is much larger than previously reported data for cells that were simply charge-discharge cycled.[17] This is because the cells spend an extended period of time at high potential every cycle which promotes more electrolyte oxidation. The cells with 0.5% PRZ + 1% DTD have the smallest charge end point capacity slippage. Again, as in Figure 7e there are no cells that give a huge advantage over control cells which means the electrolyte oxidation reactions on cathode surface are still occurring to some degree in these blended additives. Figure 9c shows the volume increase of the cells during the UHPC cycling. A clear trend in Figure 9c is that when the PBF concentration increases in cells with two additives (i.e. from 1% PBF + 1% MMDS to 2% PBF + 1% MMDS or from 1% PBF + 1% DTD to 2% PBF + 1% DTD) the amount of gas produced generated increases as well. Figure 9d shows a summary of all the impedance data for cells before and after UHPC testing. The impedance of the control cell increased dramatically during testing while the impedance of all other cells (except 2% LUTID + 1% MMDS) decreased. Cells with 1% PCN + 1% MMDS and 0.5 PRZ + 1% DTD show very small impedance after UHPC testing.

After UHPC cycling, the same cells were transferred to long-term cycling between 2.8 V and 4.5 V at 40.0°C using currents corresponding to C/2.5 (100 mA). Figure 10a shows the capacity versus cycle number for the unclamped NMC442/graphite pouch cells with comparative cells including control, PES211 and 1% MMDS. The control cells have quite poor performance during this high voltage cycling. PES211-containing cells showed good capacity retention for 250 cycles and only combinations containing 2% PBF could compete with PES211. Cells with only 1% MMDS do not perform as well as those with 2% PBF-type + 1% MMDS/DTD**.**

Figure 10b shows the discharge capacity versus cycle number for NMC442/graphite pouch cells containing different additives cycled between 3.0 V and 4.5 V at 55°C with 80 mA current (∼ C/3). These cells were cycled directly after formation and were tested without clamps. The additive combinations used for this experiment were selected based on the results of the UHPC testing (Figures 7 and 9). Comparative cells with 1% MMDS, 2% PBF alone and control electrolyte were also tested. The cells with 2% PBF and the binary additive blends showed better capacity retention than cells with 1% MMDS alone or control electrolyte. This could be a result of the Lewis base functional group (pyridine) in PBF that may be able to reduce acidic gases generated during cycling. However, since this cycling condition is very aggressive, even the best cells reached 80% capacity retention in about 220 cycles and the difference between cells with 2% PBF and 2% PBF + 1% MMDS is small.

Figure 11 shows Nyquist plots for the NMC442/graphite pouch cells before and after the long-term cycling test shown in Figure 10b. These impedance spectra were measured at 3.8 V and at 10 ± 0.1°C. The impedance of the control cells increased dramatically after cycling. By contrast, the impedance of cells with 2% PBF and the PBF-type + 1% MMDS blends stayed under control and did not increase dramatically even when cycled at a high temperature (55°C) and potential (4.5 V). The cells with 2% 3F-PBF + 1%MMDS showed a significant reduction in impedance from 310 Ω-cm^2 to 140 Ω-cm^2 in good agreement with the results in Figure 9d for UHPC cycling. This combination also showed the best capacity retention in Figure 10b.

Conclusions

Additive blends combining PBF-type molecules and MMDS or DTD were evaluated in NMC442/graphite pouch cells cycled to 4.4 V or 4.5 V. The FRA results with a special charge/hold protocol (Figures 3) demonstrated that the additive combinations are more beneficial than each component itself. The FRA data showed that the capacity retention of cells with additive combinations was excellent even when aggressive protocols were applied (Figure 3). When MMDS was added to PBF-type additives, the impedance of cells was smaller than the impedance of cells with only PBF-type additives (Figures 4 and 5). UHPC data collected in an aggressive charge/store protocol indicated that PBF-type additives helped to increase the CE of cells with additive combinations compared with those only containing MMDS. However, the charge end point capacity slippage was not dramatically improved suggesting electrolyte oxidation is still problematic. The long-term cycling results (55°C, 4.5 V) showed that cells with additive blends displayed better capacity retention than those with only with 1% MMDS or control electrolyte. At 40°C, long-term cycling results showed that cells with 1% PBF + 1% MMDS, 2% PBF + 1% MMDS or 2% PBF + 1% DTD possessed even better

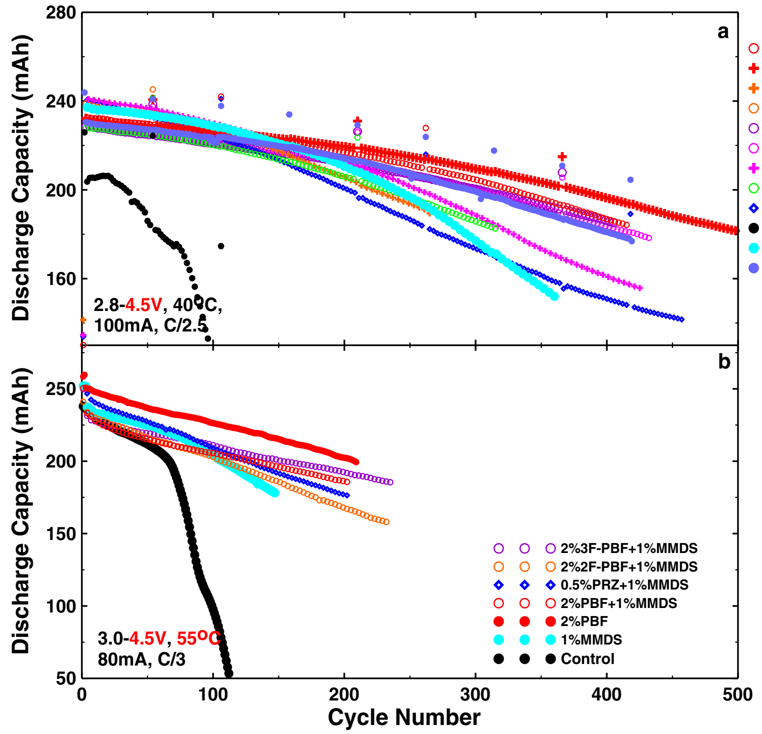

Figure 10. a) Discharge capacity versus cycle number for NMC442/graphite pouch cells cycled without clamps at C/2.5 (0.1 A) at 40.0 ± 0.1°C between 2.8 and 4.5 V with different additive blends; b) Discharge capacity vs. cycle number for NMC442/graphite pouch cells cycled without clamps at C/3 (80 mA) at 55.0 ± 0.1°C between 3.0 and 4.5 V with various PBF-type + 1%MMDS combinations.

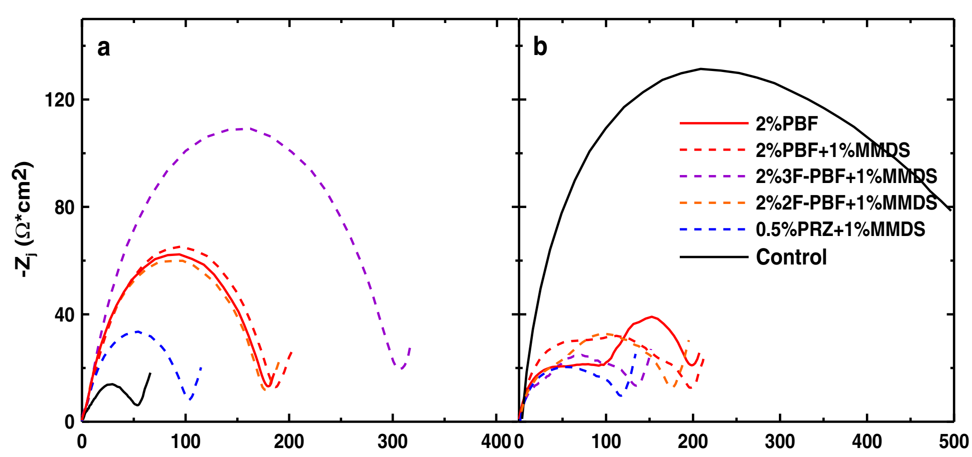

Figure 11. Impedance spectra of the cells described in Figure 10b; a) before long-term cycling; b) after long-term cycling.

capacity retention than cells with the PES211 additive combination. Therefore, as demonstrated by remarkable reductions in impedance growth of cells cycled to high potential, the PBF-type + MMDS/DTD additive combinations show potential to be high-voltage additive candidates for NMC/graphite lithium ion batteries.

As a final note, the authors believe that there are two dominant factors that govern the lifetime of NMC/graphite cells charged to high potential. There is the electrolyte oxidation that occurs and the impedance increase that it can cause. The experiments in this paper suggest that the PBF-type + MMDS/DTD additive blends are very effective at suppressing the impedance increase, but do not stop the electrolyte oxidation, as indicated by charge end point capacity slippage. Further work is required to further suppress the electrolyte oxidation.

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